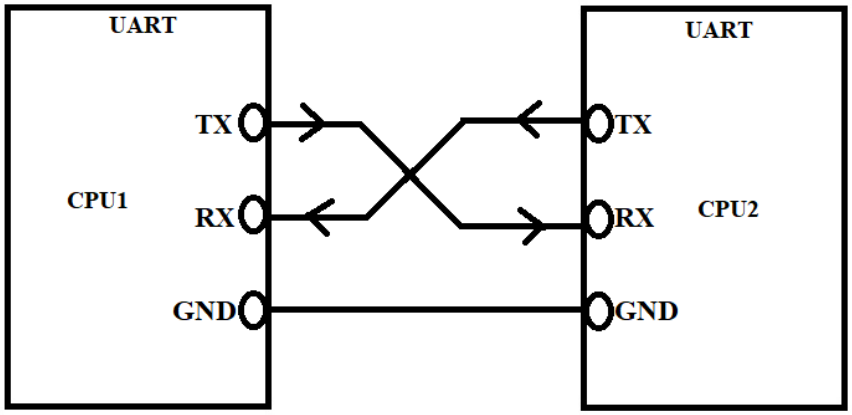
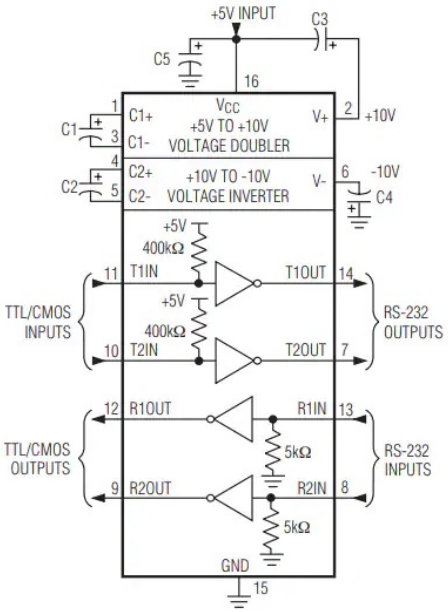
2 8051's Having OnChip UART Can be Connected as given below:

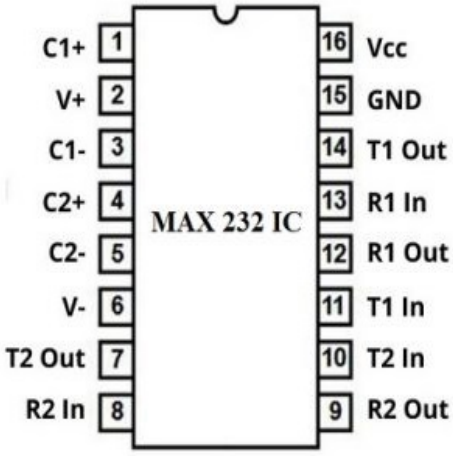


Generally, a microcontroller works at TTL of about 0 to 5V while a computer operates on the standards of RS232 like -25V to +25V.

If microcontroller wants to communicate with computer through UART communication, one voltage converter is required. i.e. MAX232

The main function of Max 232 IC is to change the signal of TTL logic level into its corresponding RS-232c level signal & then Rs-232c level signal to its corresponding TTL level signal.

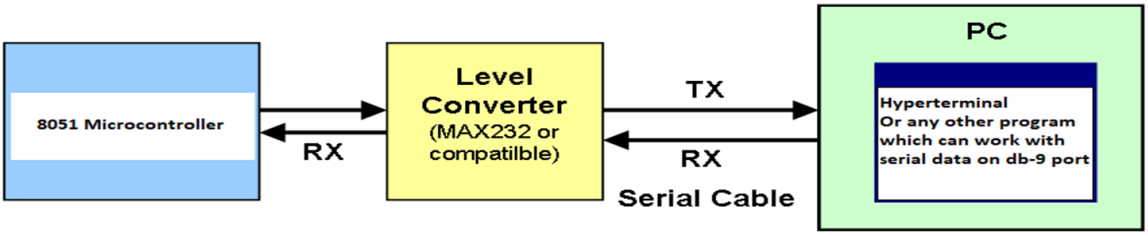






MAX232 includes three blocks.
First block is voltage doubler.
Second block will convert the voltage to +10V & -10V
Third block includes two transmitters & two receivers which convert the levels of voltage.

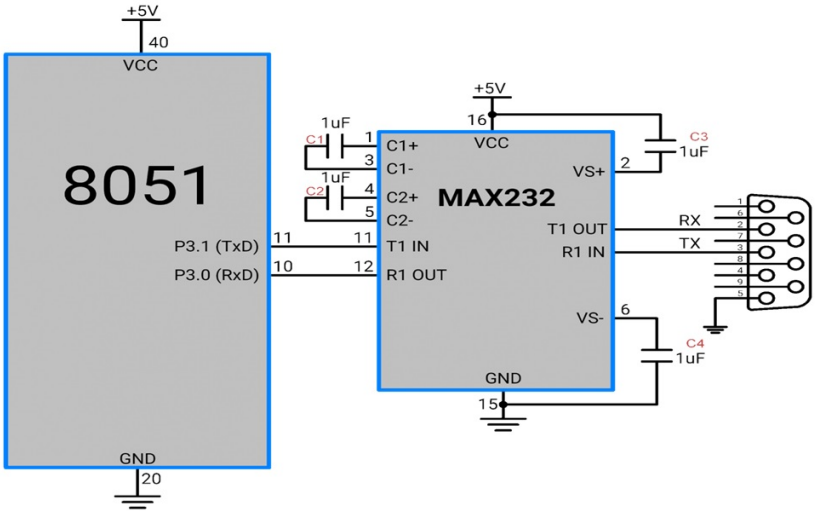
Microcontroller - PC Connection: Block Diagram

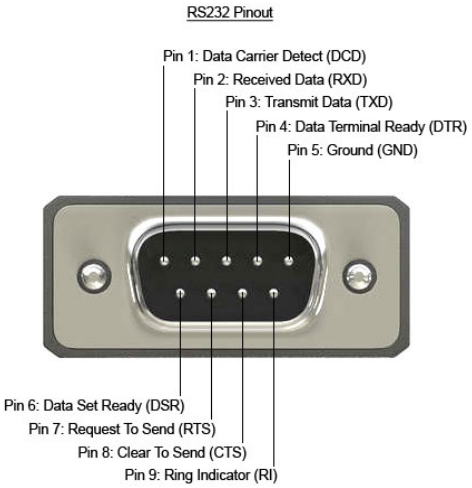


Microcontroller - PC Connection: Circuit Diagram

TTL	
LOGIC 0	0 V
LOGIC 1	5 V

RS232	
LOGIC 0	+3 to +25 V
LOGIC 1	-3 to -25 V





Detailed Connections of 8051 CPU to PC Interface:

